



**Course Number – Section: 4260-002 CRN#: 24228**

|               |                           |             |             |
|---------------|---------------------------|-------------|-------------|
| SEMESTER:     | Summer 2026               | INSTRUCTOR: | Reza Ghaeli |
| COURSE TIME:  | Thursdays 12:30 – 15:20   | ROOM:       | Anvil 809   |
| EMAIL:        | ghaelim@douglascollege.ca | TELEPHONE:  |             |
| OFFICE HOURS: | Wednesday 10:30 – 12:20   | LOCATION:   | N4304       |

*\*All times shown are in Pacific Standard Time (PST)*

### **COURSE MATERIALS REQUIRED**

All instructor materials, resources, assignments and communication such as announcements, course messages and other resources will be shared through Douglas College Blackboard Community. Unless it is specifically mentioned, all assignments should be submitted through Blackboard. It is the students' responsibility to check the announcements before coming to class. All documents as well as media, including classroom recordings, projects, and instructional material are copyrighted and are solely for the purpose of the students. Distribution of the material in any form is strictly prohibited. Specifically, recordings of the classroom instructions by students should not be shared or distributed.

### **CALENDAR COURSE DESCRIPTION**

In this course, Students will learn about emerging technologies and trends in Data Analytics. This course is divided into several modules. Each module represents a specialized body of knowledge focusing on the technical aspects of the 4V's of big data (Volume, Velocity, Variety, and Veracity) as well as policy and other aspects such as privacy and ethics. Students will also get a chance to research state-of-the-art Data Analytics in an industry of their choice. This course will provide students the required breadth to jumpstart their career in the Data Analytics field.

### **COURSE CONTENT**

- Module 1 (2 weeks): Capturing, managing and using data for decision making.
- Module 2 (3 weeks): Using tools for mining different types of data such as structured data, text data, and web data.
- Module 3 (3 weeks): Building the technology stack for Data Analytics in terms of 4 V's of big data, i.e. Volume, Velocity, Variety, and Veracity.
- Module 4 (3 weeks): Research Data Analytics in different industries.
- Module 5 (1 week): Workshop on data security issues.
- Module 6 (1 week): Workshop on ethics and privacy issues.

## LEARNING OUTCOMES

- Describe the technical requirements of managing extensive amounts of data.
- Use various tools to manipulate, map and reduce a large data set.
- Design and implement a technical solution to deal with 4 V's (Volume, Velocity, Variety, and Veracity) of big data.
- Appraise data scaling strategies such as different types of partitioning and replication in relation to different data growth and data consumption scenarios.
- Evaluate the state of Data Analytics in a chosen industry.
- Explain concepts of data security with regards to Analytics data in storage and in transmission.
- Discuss the basic principles of ethical conduct in relation to Data Analytics.
- Discuss the basic principles of data privacy in relation to Data Analytics.

## METHODS OF INSTRUCTION:

- Instructional Sessions
- Seminars
- Labs

## COMMUNICATION WITH INSTRUCTOR:

All academic related communication through emails must originate or destined from/to a valid xxxxxxxx@xxxxxx.douglascollege.ca email address. **Include your Course and Section number in the Subject line (CSIS-4260-002)** of your email. Emails originated from a different email address or without proper subject line will be disregarded due to communication management and security reasons.

**Note:** Only emails originated from your Douglas College email will be answered. Your emails should contain your Course Name and Section. Any email not from Douglas College may be lost or go to Spam Folder. Any other message to instructor through other channels including Blackboard Messages may not be checked.

## MEANS OF ASSESSMENT:

|                                |      |
|--------------------------------|------|
| Assignments, Seminars and Labs | 25%  |
| Group Project                  | 25%  |
| Quizzes                        | 15%  |
| Final Exam                     | 35%  |
| TOTAL                          | 100% |

Notes:

- If you miss more than 30% of scheduled classes, you receive [UN Grade](#) for your course grade.

## GRADING POLICY:

| Grade | Numerical Value | Achievement Level |
|-------|-----------------|-------------------|
| A+    | 4.33            | 90% to 100%       |
| A     | 4.00            | 85% to 89%        |
| A-    | 3.67            | 80% to 84%        |
| B+    | 3.33            | 77% to 79%        |
| B     | 3.00            | 73% to 76%        |
| B-    | 2.67            | 70% to 72%        |
| C+    | 2.33            | 65% to 69%        |
| C     | 2.00            | 60% to 64%        |
| C-    | 1.67            | 55% to 59%        |
| D     | 1.00            | 50% to 54%        |
| F     | 0.00            | 49% and below     |

| Grade | Numerical Value | Achievement Level                                                                                                                                                                                           |
|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UN    | 0.00`           | Students completed less than 70% of the total evaluation of the course or <u>missed more than 30% of the class where the instructor's Course Outline specifies that attendance is a course requirement.</u> |
|       |                 |                                                                                                                                                                                                             |
| W     | N/A             | Does not include GPA calculation                                                                                                                                                                            |

## REGULATIONS FOR STUDENTS:

LATE ASSIGNMENTS: Late assignments/labs will not be graded and receive an automatic zero mark except for extraordinary circumstances or prior arrangements with the instructor. Students are encouraged to keep extra copies (i.e., photocopies or file backups) of their assignments in case of data loss in the digital world.

MISSED TESTS OR FINAL EXAMINATION: Student will receive a zero mark for any missed test(s). Exceptions may be considered in cases of extraordinary circumstances such as accidents, deaths in the family, family emergencies' including sick children. It is the responsibility of the student to inform the College and/or the instructor at the earliest reasonable opportunity. Notification of the possibility of missing the test or exam must be done prior to the test or exam date/time and based on the instructor's preference might require supportive documentation where applicable.

## CLASSROOM CIVILITY AND SHARED RESPONSIBILITY:

STUDENT CONDUCT: Any student who displays disruptive or dangerous behavior will be asked to leave the classroom/lab by the instructor. Such behavior will be classified as misconduct. Reprimands and appeals will be exercised according to the [Douglas College Student Conduct policy](#).

TIMELINESS: Students are expected to be in class at the start of class. Any late student should enter the session and try to not interrupt the flow of class activity as per [Douglas College Student Conduct policy](#).

CLASS CANCELLATION: If a class is cancelled due to unforeseen circumstances, a notification will be made through Blackboard to every student enrolled in the course. It is the responsibility of students to

be proactive and to check their announcements and/or e-mail before coming to class. Every effort will be made to ensure that the notification is made as soon as possible.

ILLNESS AND OTHER UNAVOIDABLE CIRCUMSTANCES: Except in extraordinary circumstances, quizzes, tests, exam and assignment deadlines must be adhered to. If unable to attend or submit, advance notice must be provided via email at your earliest opportunity. On the email include

- Course and section number (e.g., CSIS1190-006)
- Your name and student number (e.g., Student Number 212121212)
- Late assignment or missed quiz (e.g., Missed Term Test #1)
- Brief comment (e.g., Explanation of reasoning)

Without documentation such as a doctor's letter, the instructor will discuss the most appropriate course of action that will lead to fair evaluation of your overall learning in the course. Students must use their Douglas College email account to communicate with the instructor and communication must be in English.

PREPARATION, ATTENDANCE AND PARTICIPATION: Attendance will be taken on a regular basis. The method of delivery includes classroom discussion and lab exercises; and students need to be present to participate and to learn.

STUDENT EFFORT: In addition to the scheduled times for classes and labs, students are expected to spend at least 10 hours a week on this course. If you are consistently spending more time than this, consider speaking with your instructor or reaching out to the [Accessibility Centre](#) for assistance.

## Course Schedule

| WEEK #  | DATES      | WEEKLY TOPICS AND ACTIVITIES                                                                                                 | READINGS AND ASSIGNMENT DUE DATES |
|---------|------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Week 01 | 2026-05-07 | Course Objectives and Structure<br>Course Introduction<br><b>Introduction +</b><br>Big Data for decision making.             |                                   |
| Week 02 | 2026-05-14 | Cloud Computing and Big Data<br>The New Data Landscape : Introduction to Big Data                                            |                                   |
| Week 03 | 2026-05-21 | Data Mining,<br>The ACE Framework for Data Decisions<br>Data Sources, Data Formats<br>Cleaning and Preparing Data Processing | Quiz 1 (5%)                       |
| Week 04 | 2026-05-28 | Distributed File Systems,<br>Hadoop and Its Distributed File System<br>Research Analytics in different industries            |                                   |
| Week 05 | 2026-06-04 | Big Data Tools and Technologies                                                                                              | Project Proposal Review           |
| Week 06 | 2026-06-11 | Big Data Tools and Technologies                                                                                              | Quiz 2 (5%)                       |
| Week 07 | 2026-06-18 | Big Data Tools and Technologies                                                                                              |                                   |
| Week 08 | 2026-06-25 | Big Data Tools and Technologies                                                                                              |                                   |
| Week 09 | 2026-07-02 | Applications, Iterative Processing, and Data Streaming                                                                       | Click or tap here to enter text.  |
| Week 10 | 2026-07-09 | Guest Speaker Session                                                                                                        | Quiz 3 (5%)                       |
| Week 11 | 2026-07-16 | Industry Seminar Series                                                                                                      | TBA                               |
| Week 12 | 2026-07-23 | Ethics & Privacy                                                                                                             | TBA                               |
| Week 13 | 2026-07-30 | <b>Project Presentations</b>                                                                                                 | TBA                               |
|         |            | <b>FINAL EXAMINATION PERIOD</b>                                                                                              | TBA                               |

Note that the exact course content and schedule of topics shown above may be altered at the instructor's discretion. *Exam dates will NOT be changed to accommodate your travel plan. You must not make any travel arrangement on or before your exam.*

This following schedule is tentative and subject to change, as per the College policy. Please do not make any travel arrangements during the final examination period – final exam scheduling is beyond the instructor's control. Please see the Registrar's office immediately with any conflict(s).

**LINKS TO IMPORTANT INFORMATION AVAILABLE ON COLLEGE WEBSITE:**

1. [Minimum technical requirements for taking courses online at Douglas College](#)
2. [Technical support information for students on the College website](#)
3. [Academic Integrity Policy \(Douglas College Educational Policy\)](#)

Plagiarism and Cheating:

The use and/or reference of any/all websites (e.g. coursehero.com or similar) which host copies of Douglas College course work assessments such as but not limited to Quizzes, assignments, midterms, labs, exams, practical work, etc. constitutes plagiarism.

4. [Course transferability](#)
5. [COVID-19 safety and guidance](#)
6. [Dates and Deadlines](#)
7. [Bookstore](#)
8. [Accessibility Services](#) – Carrie Keen for CBA Students
9. [Library](#)

!!!\*\*\* WISH YOU ALL THE VERY BEST FOR THIS COURSE \*\*\*!!!